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SDLC

ASSIGNMENT – 1

Assignment 1: SDLC Overview - Create a one-page infographic that outlines the SDLC phases (Requirements, Design, Implementation, Testing, Deployment), highlighting the importance of each phase and how they interconnect.

Software Development Life Cycle (SDLC) Overview



Requirements

Purpose: Understand what
Activities: Stakeholder meetings, documentation of
Importance: Sets a clear foundation and avoids scope creep



Design

Purpose: Plan the structure and components of the system.
Activities: Create diagrams (UML, data flow), choose technology
Importance: Guides developers and ensures technical feasibility



Implementation

Purpose: Convert design into actual code
Activities: Programming, code reviews
Importance: Builds the functional system



Testing

Purpose: Ensure the system meets requirements and is bug-free
Activities: Unit testing, integration testing, user acceptance testing
Importance: Validates quality and reliability



Purpose: Make the system available
Activities: Install software, train users, monitor performance

How They Interconnect:

Each phase builds upon the **previous** one. For example, design follows requirements, and testing verifies implementation. A failure in one stage can impact all ers.

A Software Development Life Cycle (SDLC) Overview

1 Requirements

- Purpose: Understand what the users need.
- Activities: Stakeholders meetings, documentation.
- Importance: Sets a clear foundation and avoids creep.

2 Design

- Purpose: Plan the structure and components of the system.
- Activities: Create diagrams (UML, data flow), select technology.
- Importance: Guides developers and ensures feasibility.

3 Implementation

- Purpose: Convert design into actual code.

- Activities: Programming, code reviews
- Importance: Builds the functional system

4 Testing

- Purpose: Ensure the system meets requirements and is bug-free
- Activities: Unit testing, integration testing, user acceptance testing
- Importance: Validates quality and reliability.

5 Deployment

- Purpose: Make the system available for users
- Activities: Install software, train users, monitor performance
- Importance: Delivers value to the end-user.

How they interconnect: Each phase builds upon the previous one. For example, design follows requirements, and testing verifies the implementation. A failure in one stage can impact all others.